

Skill 10

Task1: To Parse Combined Redirection, Duplicate File Descriptors, Merge Streams, Verify Ordering, Handle Edge Cases, Test Integrated Outputs.

Code:

```
#include <stdio.h>
```

```
#include <fcntl.h>
```

```
#include <unistd.h>
```

```
int main()
```

```
{
```

```
    int fd = open("combined.txt", O_WRONLY | O_CREAT |  
O_TRUNC, 0644);
```

```
    if (fd < 0)
```

```
        return 1;
```

```
    dup2(fd, STDOUT_FILENO);
```

```
    dup2(STDOUT_FILENO, STDERR_FILENO);
```

```
    close(fd);
```

```
    printf("Standard output message.\n");
```

```
    fprintf(stderr, "Standard error message.\n");
```

```
    return 0;
```

```
}
```

Output:

```
ashwithreddy_earla@AshwithReddy:~$ nano Skill10.c
ashwithreddy_earla@AshwithReddy:~$ gcc Skill10.c -o Skill10
ashwithreddy_earla@AshwithReddy:~$ ./Skill10
ashwithreddy_earla@AshwithReddy:~$ cat combined.txt
Standard error message.
Standard output message.
ashwithreddy_earla@AshwithReddy:~$ |
```

Task2: To Parse Complex Commands, Combine Pipes and Redirections, Resolve Execution Order, Validate Syntax, Generate Execution Plans, Test Scenarios.

Code:

```
#include <stdio.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>


int main()
{
    int p[2];

    pipe(p);

    if (fork() == 0)
    {
        dup2(p[1], STDOUT_FILENO);

        close(p[0]);
```

```

        close(p[1]);

        execlp("ls", "ls", NULL);

        return 1;
    }

    if (fork() == 0)
    {
        dup2(p[0], STDIN_FILENO);

        close(p[0]);
        close(p[1]);

        execlp("grep", "grep", ".c", NULL);

        return 1;
    }
    close(p[0]);
    close(p[1]);
    wait(NULL);
    wait(NULL);
    return 0;
}

```

OUTPUT:

```

Standard output message:
ashwithreddy_earla@AshwithReddy:~$ nano Skill10.1.c
ashwithreddy_earla@AshwithReddy:~$ gcc Skill10.1.c -o Skill10.1
ashwithreddy_earla@AshwithReddy:~$ ./Skill10.1
Experiment10.c
Experiment2.c
Experiment5.c
Experiment6server.c
Experiment7.1.c
Experiment8.1.c
Experiment8.2.c
Experiment8.c
OSSP.c
Skill10.1.c
Skill10.c
Skill5.c
Skill8.1.c
Skill8.2.c
Skill8.3.c
Skill8.c
Skill9.1.c
Skill9.2.c

```

Task3: To Detect Background Jobs, Launch Non-Blocking Processes, Return Prompt Immediately, Store Job Information, Monitor Execution, Test Background Tasks.

Code:

```

#include <stdio.h>

#include <unistd.h>

#include <sys/types.h>

#include <sys/wait.h>


int main()
{
    pid_t pid = fork();

    if (pid == 0)

```

```
{
    for (int i = 1; i <= 5; i++)
    {
        printf("Background job: %d\n", i);
        sleep(1);
    }

    return 0;
}

printf("[1] Background job started with PID %d\n", pid);
printf("Parent continues immediately.\n");

waitpid(pid, NULL, 0);

printf("Background job completed.\n");

return 0;
}
```

OUTPUT :

```
ashwithreddy_earla@AshwithReddy:~$ nano Skill10.2.c
ashwithreddy_earla@AshwithReddy:~$ gcc Skill10.2.c -o Skill10.2
ashwithreddy_earla@AshwithReddy:~$ ./Skill10.2
[1] Background job started with PID 709
Parent continues immediately.
Background job: 1
Background job: 2
Background job: 3
Background job: 4
Background job: 5
Background job completed.
ashwithreddy_earla@AshwithReddy:~$ |
```

Task4: To Create Job Table, Store Process Groups, Track States, Update Status, Remove Completed Jobs, Validate Job Records.

Code:

```
#include <stdio.h>
```

```
#include <sys/types.h>
```

```
typedef struct
```

```
{
```

```
    int id;
```

```
    pid_t pid;
```

```
    char state[20];
```

```
} Job;
```

```
int main()
```

```
{
```

```
    Job jobs[3] = {
```

```
        {1, 4217, "Running"},
```

```

        {2, 4218, "Stopped"},

        {3, 4219, "Running"}

    };

    printf("JOB TABLE\n");

    printf("ID\tPID\tSTATE\n");

    for (int i = 0; i < 3; i++)

        printf("%d\t%d\t%s\n",

            jobs[i].id,

            jobs[i].pid,

            jobs[i].state);

    return 0;

}

```

OUTPUT:

```

ashwithreddy_earla@AshwithReddy:~$ nano Skill10.3.c
ashwithreddy_earla@AshwithReddy:~$ gcc Skill10.3.c -o Skill10.3
ashwithreddy_earla@AshwithReddy:~$ ./Skill10.3
JOB TABLE
ID      PID      STATE
1       4217     Running
2       4218     Stopped
3       4219     Running
ashwithreddy_earla@AshwithReddy:~$ |

```